

Task 1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;
    source = open("source.txt", O_RDONLY);
    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }
    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }
    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        write(destination, buffer, bytesRead);
    }
    close(source);
    close(destination);
    printf("File copied successfully.\n");
    return 0;
}
```

Output:

```
rohita@LaxmiDurga:~$ nano new12.c
rohita@LaxmiDurga:~$ echo echo "This is my source file." > source.txt
rohita@LaxmiDurga:~$ gcc new12.c -o new12
rohita@LaxmiDurga:~$ ./new12
File copied successfully.
rohita@LaxmiDurga:~$
```

Observation: The program successfully copies the contents from the source file to the destination file using `open()`, `read()`, `write()`, `close()`.

Task2: Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

